PREVAIL
Predictive Response for Emergency Volume Assessment in Incident Locations

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Abstract

Extreme weather conditions pose a critical threat to utility infrastructure, yet utility companies must often rely on reactive, ”wait-and-see” scheduling that results in inefficient crew deployment and prolonged restoration times. While previous studies have successfully used machine learning to predict the probability of outage occurrence during adverse weather events, these binary classification models fail to quantify the specific resource allocation required for timely restoration of service. To address this operational gap, this project introduces PREVAIL: Predictive Response for Emergency Volume Assessment in Incident Locations, a framework designed to directly translate meteorological forecasts into actionable workforce logistics. Rather than predicting if or where an outage will occur, PREVAIL focuses strictly on resource quantification, forecasting the precise crew sizes required for restoration over a weekly planning window. By engineering a novel spatio-temporal linkage between historical outage logs and crew dispatch records using a ZIP code proxy, we constructed a training dataset of over 1,500 verified adverse weather-related responses. We utilize this data to train an ensemble of machine learning models to enable proactive crew staging, thereby reducing standby costs while enhancing grid reliability.

Website: https://angela139.github.io/prevail-dashboard/
Code: https://github.com/adityasurap/PREVAIL

Please note that due to a data privacy agreement with SDG&E, we are not able to make this repository public. Please contact Aditya Surapaneni at the corresponding email address listed above to receive access.

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1 Introduction

1.1 The Problem

Extreme weather events are the single largest cause of power outages in the United States, and cost an estimated $20 to $55 billion annually. For utility operators like San Diego Gas & Electric (SDG&E), the challenge is not merely in preventing outages, but also in responding to them efficiently when they do occur. Current operational protocols often rely on static heuristics or reactive dispatching. In other words, utilities often wait for a customer to report an outage event before a crew can be assigned and dispatched. This ”wait-and-see” approach leads to significant inefficiencies. If an operator underestimates the severity of a storm, crews are dispatched at the last minute which delays restoration. Similarly, if they overestimate, contract crews sit idle on standby which increases costs for homes and businesses. The objective of this project is to transition this decision-making process from reactive to proactive. By predicting the specific crew size required for impending weather-driven outages, we aim to optimize resource allocation before the weather event even makes landfall.

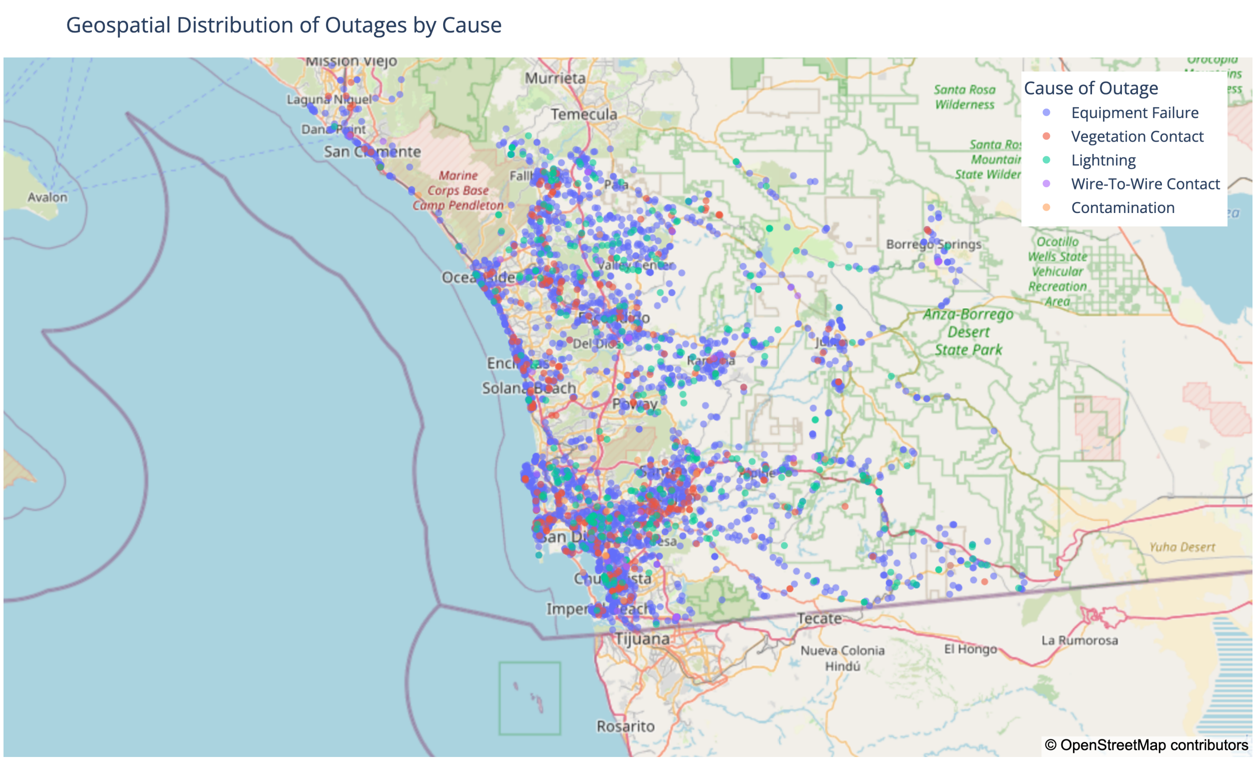

Figure 1: Geospatial distribution of historical power outages across the SDG&E service territory, color-coded by primary cause. The density and variety of these events illustrate the logistical complexity of coordinating a proactive emergency response.

1.2 Literature Review and Related Work

The application of machine learning to enhance power grid reliability is a well-established field. However, prior work has focused predominantly on occurrence prediction and duration estimation rather than repair and resource quantification.

Occurrence/Duration Prediction: Extensive research has been conducted on predicting the spatial distribution of outages. The most prominent framework is the Hurricane Outage Prediction Model, which utilizes Generalized Linear Models (GLMs) and Random Forests to estimate the probability of infrastructure failure based on wind speed, soil saturation, and vegetation cover. (Guikema et al., 2014) Subsequent improvements by Wanik et al. (2015) integrated dynamic weather variables to improve accuracy in the Northeastern United States. These models effectively identify high-risk zones and are fundamentally designed to predict the likelihood or counts of failure events.

The Resource Gap: A parallel body of research focuses on the Estimated Time of Restoration (ETR) by predicting how long customers will be without power. (Nateghi et al., 2011) However, there remains a significant gap in the literature regarding the input side of the restoration equation: the specific labor resources required to restore service. Current models often treat crew availability as a static input parameter for simulation rather than a dynamic target variable to be predicted. Existing literature rarely integrates crew dispatch logs into the predictive pipeline, largely due to the difficulty of linking disparate data sources. Our project addresses this gap by explicitly modeling the target variable `crew_size`, shifting the analytical focus away from traditional binary outage classification, and directly introducing a regression framework that informs proactive operational logistics.

1.3 Data Description

To address the challenge of quantifying restoration resources, this project integrates three distinct, high-volume datasets provided by SDG&E, covering the period from 2019 to 2025. The unification of these disparate data sources, including operational logs, dispatch records, and meteorological sensor data, is the fundamental enabler of our predictive framework.

1. Historical Outage Data: The core dataset is sourced from the Outage Management System (OMS), containing over 460,000 historical power outage records. This dataset provides critical baseline features such as outage start time, duration, cause code (e.g., ”Wind”, ”Vegetation”), and precise GPS coordinates of the failed asset. While OMS data allows us to identify historical incident locations, it notably lacks information on the human response effort, necessitating the integration of secondary datasets.

2. Resource Allocation Logs: To model our target variable, `crew_size`, we utilized the Scheduling Outage Response Team (SORT) database. Unlike the automated sensors of the OMS, the SORT dataset represents the human dimension of grid reliability. It logs the number of personnel dispatched, the type of equipment requested, and the service window. A major challenge in this domain is that SORT records are aggregated by ZIP code rather than precise GPS coordinates. To resolve this, we engineered a spatial proxy that links

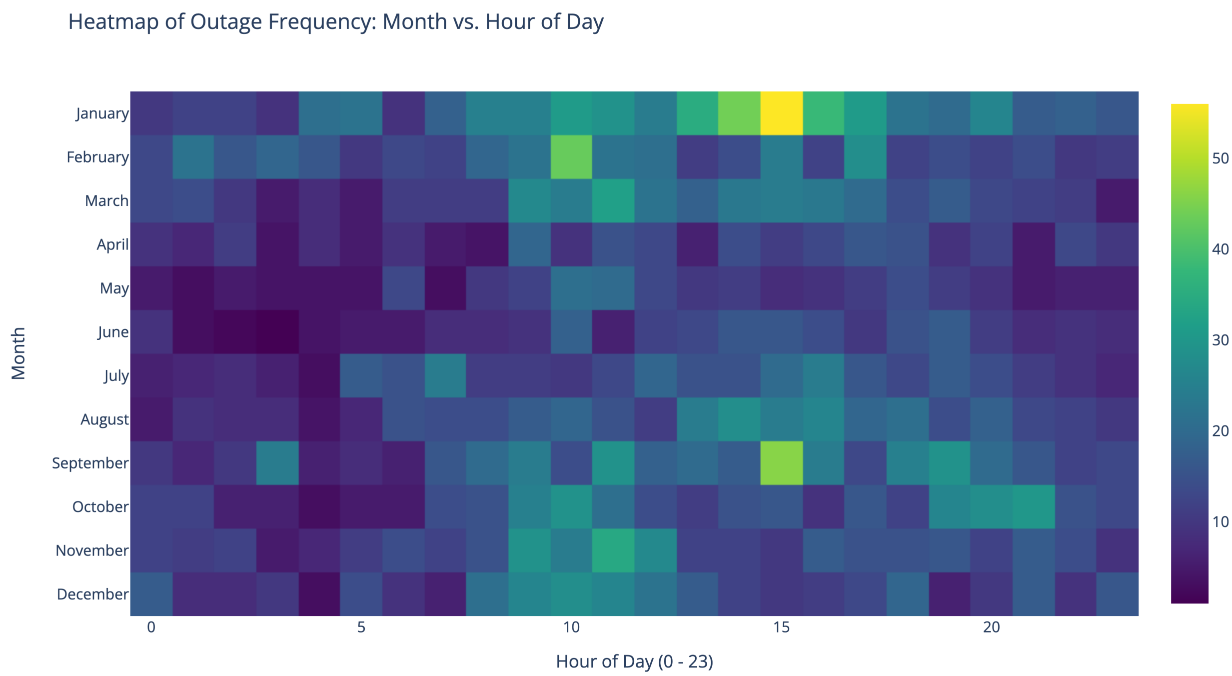

Figure 2: Heatmap of historical outage frequency across the SDG&E service territory, categorized by month and hour of day. The distribution reveals clear temporal peaks, highlighting the cyclical nature of grid stress and the importance of temporal features in the PREVAIL model.

SORT records to OMS outages within a shared ZIP code and twelve-hour response window, successfully recovering over 1,500 labeled instances of high-impact storm responses. This dataset allows our model to distinguish between simple outages requiring only one crew member and more complex events requiring larger crew sizes of three or more.

3. Meteorological Sensor Data: The predictive engine of PREVAIL is driven by high-resolution weather data collected from a proprietary network of weather stations across the San Diego region. The raw dataset contains over 75 million hourly readings from 2019 to 2025. To manage this volume and align it with our modeling objectives, we isolated the meteorological conditions corresponding directly to our recorded dispatch events. Key features include *Wind Gust* (MPH), *Sustained Wind Speed* (MPH), *Air Temperature* (°F), and *Relative Humidity* (%). These variables were selected because they act as direct physical proxies for infrastructure stress; for example, high wind gusts combined with low humidity are strong indicators of vegetation-related line faults and fire risk. By synthesizing these three sources, we transform raw logs into a supervised learning problem: mapping atmospheric conditions directly to the required human response.

2 Methods

2.1 Dataset Preparation and Transformation

Before constructing our predictive models, we developed a master dataset by synthesizing hourly weather telemetry with historical dispatch logs. Our primary objective was to capture granular atmospheric fluctuations while maintaining strict temporal separation to prevent data leakage. We initially focused on data from 2021 to the present, ensuring all temporal features were normalized to UTC. To maximize data density, we imputed missing outage start or end times using the recorded duration whenever at least one timestamp was available.

To reduce dimensionality while eliminating redundancy, we structured the dataset around unique `outage_id` and `station_id` pairs. This allowed us to preserve the essential characteristics of each outage while accounting for variance across multiple recording stations. We aggregated numeric columns by calculating their mean values, with the exception of the `outage_duration` feature, where we recorded the maximum value to account for the most severe service disruptions. For categorical and non-numeric features, we utilized the first available non-null observation.

We then merged the weather telemetry, which contains hourly logs of all jurisdictional stations, with this cleaned outage foundation. We filtered the weather data to isolate only those observations that fell within the specific active windows of recorded outages. From this merged set, we calculated the localized averages for all relevant meteorological metrics.

To capture the impact of volatile conditions, we engineered binary indicators for extreme weather events. We established an 85th percentile threshold for temperature, wind speed, and wind gusts, and a 10th percentile threshold for minimum humidity to flag high-risk states. Additionally, we performed keyword parsing on descriptive text columns, identifying terms such as ”wind,” ”storm,” ”heat,” and ”rain” to confirm if an outage was officially classified as a weather-driven event.

One of the most predictive components of our dataset involved the engineering of ”lag” features to provide the model with historical atmospheric context. By mapping each outage back to its corresponding station in the original hourly weather logs, we generated rolling features for one, six, and twenty-four hours prior to the incident. These lags allow the model to interpret not just the conditions at the moment of failure, but the cumulative environmental stress preceding it. The resulting panel dataset, cleaned to minimize null values, serves as the robust analytical input for our modeling pipeline.

2.2 Feature Engineering for Resource Quantification

To capture the physical drivers that dictate the complexity of grid repairs and the required workforce, we engineered a set of specialized interaction features. Rather than simply predicting infrastructure failure, these features proxy the severity of the damage, which directly correlates to our target variable, `crew_size`. Our analysis highlighted the following key predictors.

- **temp_delta_1d:** Represents the absolute change in temperature from the previous day. This feature serves as a proxy for the rapid onset of thermal stress on grid components, which often leads to complex material failures requiring specialized or larger repair crews.
- **wind_rolling_7d_max:** The maximum wind speed observed over the preceding week. This serves as a baseline for recent environmental stress, indicating weakened infrastructure (such as stressed vegetation or strained poles) that might result in compounded damage during a storm.
- **wind_x_temp:** An interaction term combining high wind and high heat. This feature proxies the risk of "phase-to-phase" shorts, where high temperatures cause power lines to sag, known as thermal expansion, while high winds cause them to sway, known as galloping. This combination often results in catastrophic conductor contact, necessitating extensive personnel for safe restoration.

Additionally, early iterations revealed that treating each H3 hexagon as an isolated unit ignored the spatial autocorrelation inherent in extreme weather systems. Widespread regional damage stretches utility resources and often necessitates larger, coordinated crew deployments. To capture this, we introduced spatial lag features, including `neighbor_outage_lag1`, which represents the sum of outages in the six surrounding hexes from the previous day, and `neighbor_wind_mean`, which represents the average wind speed of neighboring hexes.

2.3 Predicting Crew Size

With our unified dataset established, we focused on building a regression pipeline capable of predicting the exact `crew_size` needed for a localized outage. To ensure both mathematical robustness and real-world operational accuracy, we developed and refined this predictive engine across three distinct stages.

2.3.1 Baseline: LASSO Regression

We established our initial baseline using a LASSO regression model. Since our engineered dataset contained over 200 highly correlated weather, infrastructure, lag, and rolling variables, we needed a robust way to handle multicollinearity without overfitting. The L1 regularization of LASSO automatically performed feature selection by shrinking the coefficients of less important, noisy variables to exactly zero. This provided us with a clean, interpretable linear baseline that achieved a Mean Absolute Error (MAE) of 1.02 crew members.

2.3.2 Refinement: XGBoost with a Poisson Objective

While LASSO effectively isolated the most important features, the relationship between extreme weather and infrastructure damage is inherently non-linear. To capture these complex interactions, we transitioned to a Gradient Boosting Regressor, specifically XGBoost.

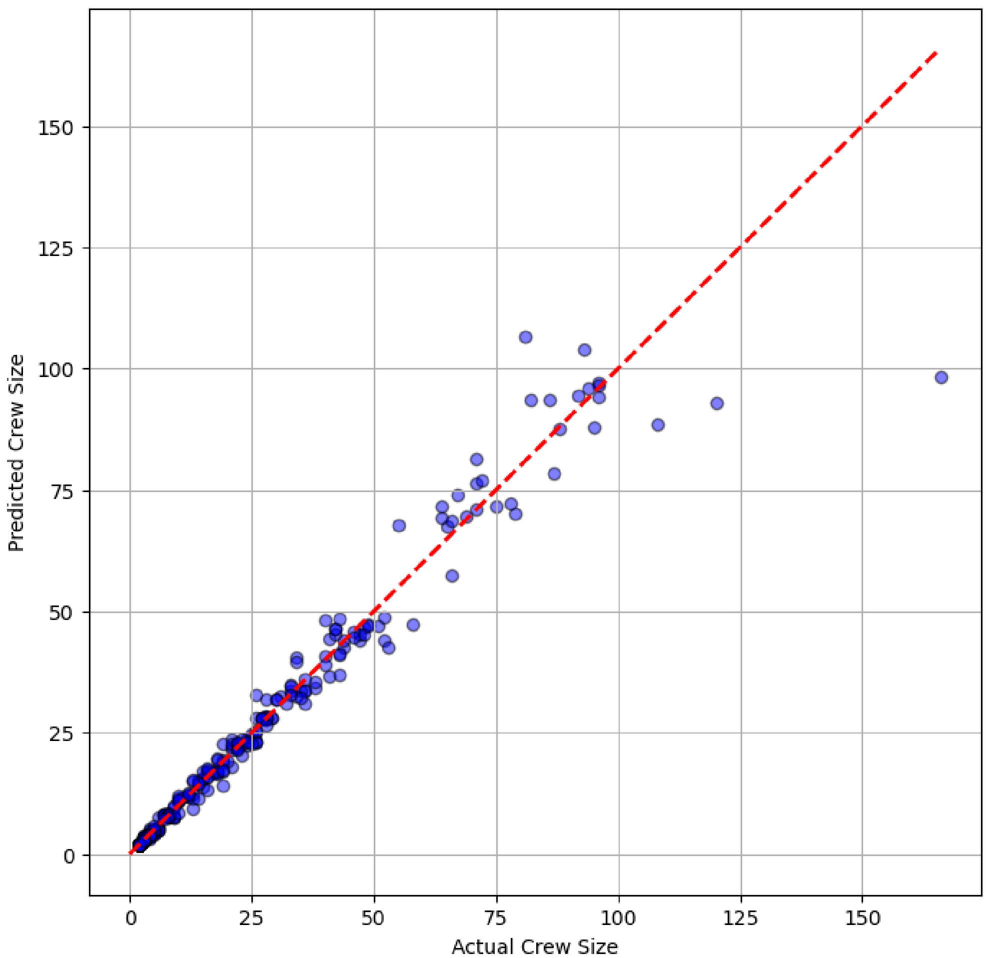

Figure 3: Regression parity plot for the LASSO model. The results demonstrate high precision for standard, low-volume crew dispatches, while showing increased variance and a tendency to underestimate for rare, high-capacity emergency events.

Crucially, we configured the model using a **Poisson objective function** rather than standard squared error. Standard regression models assume a continuous, normally distributed target, which can lead to physically impossible negative predictions or poor handling of extreme outliers. Because `crew_size` is a discrete count variable (distinct integers like 1, 2, or 4), the Poisson distribution naturally fit the zero-bounded, right-skewed shape of our dispatch data. This explicitly trained the model to understand the discrete nature of human resource allocation, improving our ability to predict outliers while slightly reducing our MAE to 0.99.

2.3.3 Final Architecture: Stacking Ensemble

To maximize predictive power, our final architecture utilized a Stacking Ensemble. We fed the predictions from both the linear LASSO model and the non-linear XGBoost model into a final meta-learner. This architecture allowed the ensemble to smooth out the inherent biases of the individual base models, leveraging LASSO's stability for standard weather conditions and XGBoost's flexibility for extreme, compounding storm events.

This combination yielded our best overall results: a MAE of 0.89 crew members, and an operational accuracy of more than 70%. In practical terms, an MAE of 0.89 means our forecasts deviate from the actual required dispatch by less than one person on average. Furthermore, achieving over 70% operational accuracy indicates that nearly three-quarters of our predictions fall safely within a strict one-person tolerance, giving dispatchers a highly reliable baseline that requires minimal manual adjustment.

2.4 Dashboard Architecture and Integration

To bridge the gap between our machine learning pipeline and the utility control room, we developed the final PREVAIL framework as an interactive web application. Rather than requiring dispatchers to interact with raw code or static reports, the dashboard provides a visual, intuitive interface designed specifically for proactive logistical planning.

The integration between the predictive model and the front-end application relies on a streamlined data handoff. First, the trained Stacking Ensemble runs inference on incoming high-frequency weather forecasts. The model generates specific `crew_size` predictions for each localized spatial grid. These predictions, alongside their corresponding H3 hex codes, mapped ZIP codes, and weather severity indicators, are exported and structured into a standardized CSV format.

The front-end dashboard was engineered using Next.js, a React-based framework selected for its fast rendering performance and scalable architecture. The Next.js application ingests the model's CSV outputs and translates the data into an interactive geospatial interface. By integrating mapping libraries, the dashboard visualizes the predicted environmental stress across the San Diego region and displays the exact recommended crew allocations for specific depots.

3 Results

The primary deliverable of this project is not merely a trained algorithm, but a fully functional, interactive web application known as the PREVAIL dashboard. By translating our Stacking Ensemble's mathematical outputs into a visual interface, we successfully connect raw data with proactive utility management. The dashboard allows dispatchers to dynamically explore historical weather events, visualize the exact workforce requirements that would have been necessary across the San Diego region. By demonstrating high accuracy on past events, the dashboard serves as a validated framework for future live deployment.

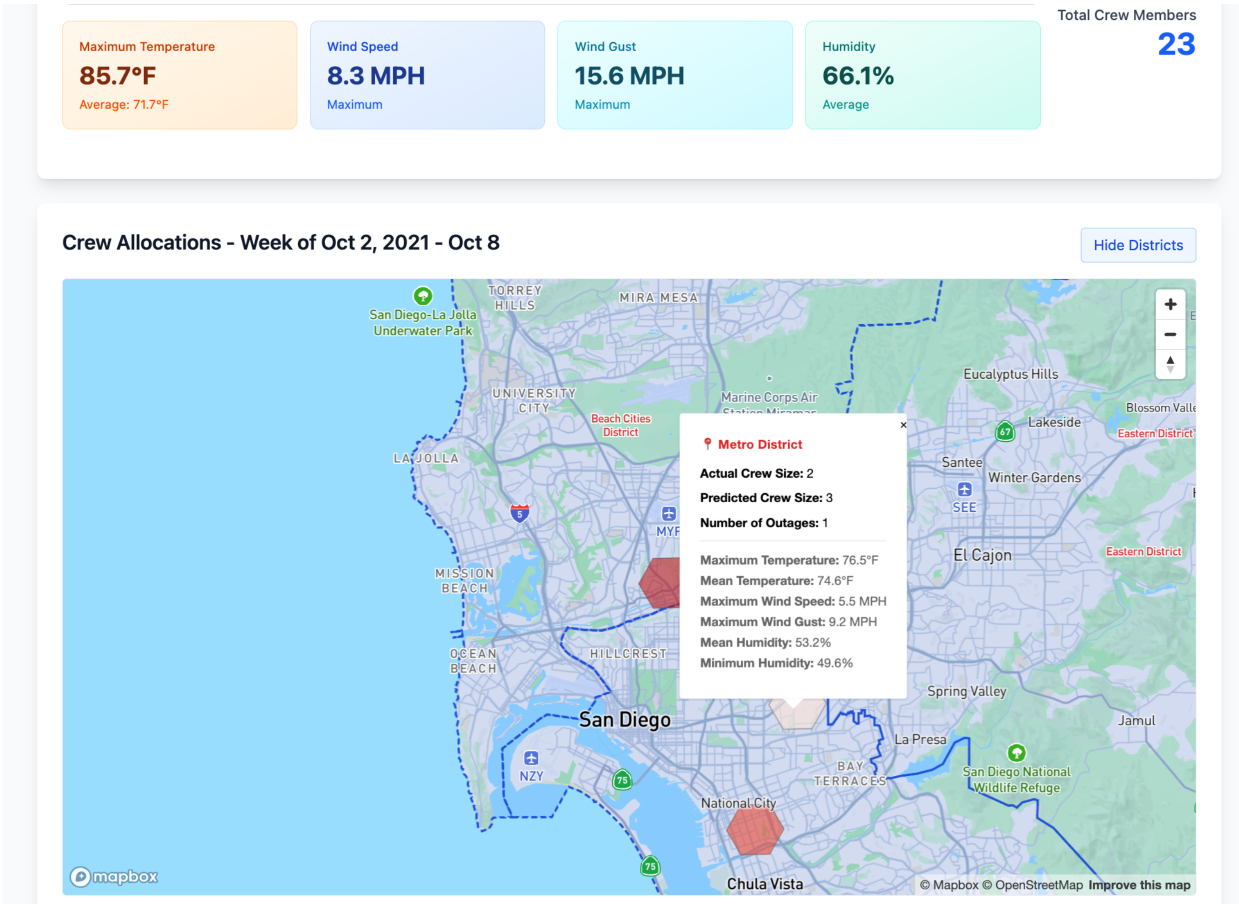

Figure 4: The PREVAIL dashboard interface, displaying weekly weather aggregates and an interactive, color-coded geospatial H3 hex map indicating predicted crew requirements.

3.1 Top-Level Analytics and Navigation

The user experience is anchored by a weekly selection tool, allowing operators to select historical time frames (spanning 2021 to 2024) to retrospectively analyze grid conditions, and validate the model's predictions against past extreme weather events. Once a week is selected, the dashboard immediately updates a top-level analytics bar. This summary provides dispatchers with an instant snapshot of the week's environmental stress, displaying the maximum and average temperatures, maximum wind speed and wind gust, and average humidity.

Most importantly, this section aggregates the model's predictions to display the *total* expected number of crew members required to handle all modeled outages for that week. This historical validation demonstrates to utility leadership how the tool successfully provides a macro-level view of necessary resource scaling.

3.2 Geospatial Visualization

Below the summary metrics lies the core visual component: an interactive map of San Diego. Rather than plotting messy, overlapping points, the map is cleanly overlaid with our resolution 7 H3 hexagons, isolating the specific areas where weather-driven outages occurred during the selected historical week.

These hexagons are dynamically color-coded based on the model's `crew_size` predictions. A gradient from yellow to red visually communicates severity: yellower hexes indicate a standard, low-resource response, while redder hexes instantly highlight complex, high-resource disaster zones.

Operators can click on any individual hexagon to pull up a detailed tooltip containing specific, localized metrics. For example, clicking on a high-risk hex in the Metro District provides the following explicit breakdown.

- **Deployment Needs:** The Predicted Crew Size (e.g., 3 members) and the historical Number of Outages, alongside the Actual Crew Size, providing instant visual validation of the model's performance.
- **Meteorological Drivers:** The localized Maximum and Mean Temperatures (e.g., 76.5°F and 74.6°F), Maximum Wind Speed and Wind Gust (e.g., 5.5 MPH and 9.2 MPH), and Mean and Minimum Humidity (e.g., 53.2% and 49.6%).

To further assist with regional planning, users can also toggle utility district boundaries on or off across the map interface.

3.3 Actionable Allocation Cards

To ensure the dashboard remains highly actionable for dispatchers, the visual map is supported by a series of "Crew Allocation Details" cards located directly beneath it. Each card corresponds directly to an active hexagon on the map. Clicking on any one of the cards highlights the corresponding hexagon while displaying the tooltip.

These cards serve as simulated marching orders for the control room. Each entry provides the exact geographic center of the hex (Latitude and Longitude) along with its unique identifier. Crucially, the cards explicitly display the required human logistics for that specific location. For instance, detailing a need for 9 crew members to handle 3 distinct outages in an area experiencing 71.8°F heat and 74.6% humidity.

By presenting both the visual heatmap and the concrete, granular allocation numbers side-by-side and validating them against real outage logs, the dashboard successfully fulfills the project's primary objective: demonstrating a deployable framework that empowers operators to proactively stage the right number of crews in the exact right locations before a future storm hits.

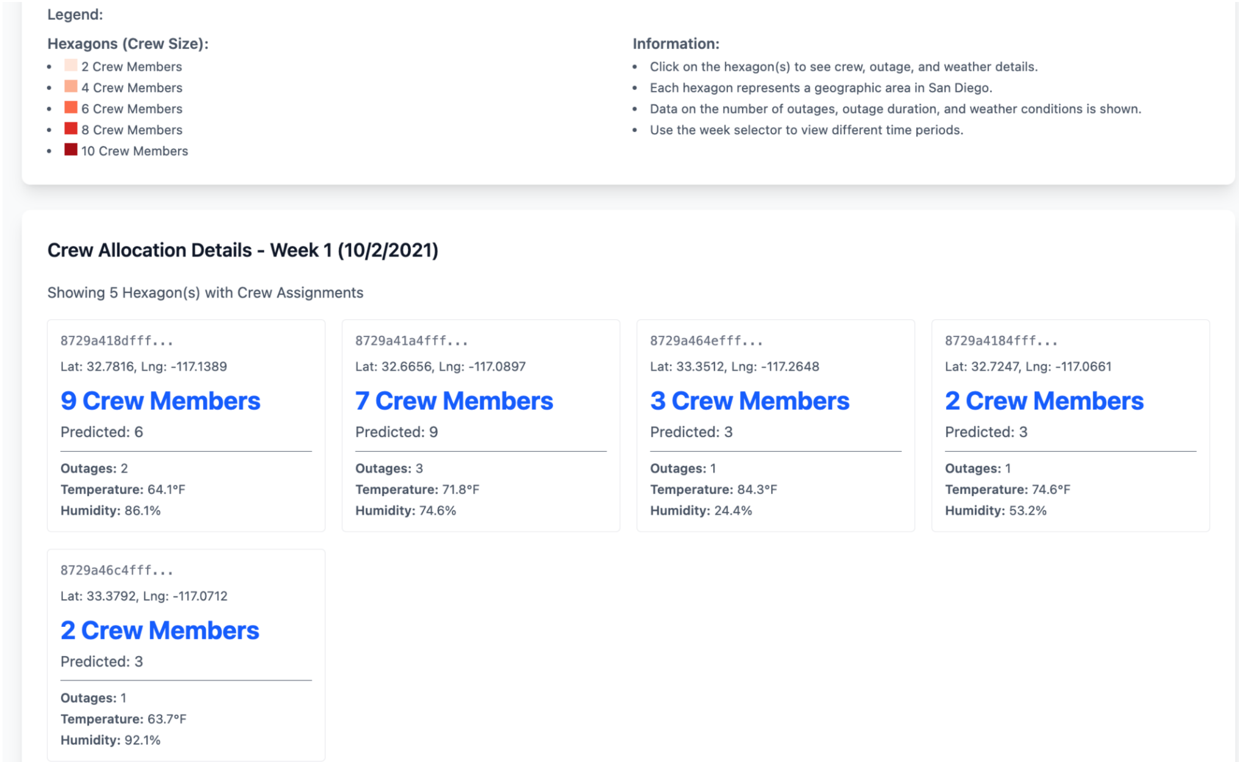

Figure 5: Detailed Crew Allocation cards, providing granular geographic coordinates, outage counts, and localized meteorological conditions to guide specific personnel staging decisions.

4 Discussion

The PREVAIL framework demonstrates that translating meteorological telemetry into actionable workforce logistics is not only mathematically viable but operationally superior to traditional ”wait-and-see” dispatching. However, while the stacking ensemble achieves high accuracy within a ± 1 crew member tolerance, several factors influence the remaining error variance, as well as the current scope of the dashboard.

4.1 Limitations

Despite the model’s strong performance, certain constraints within the data and operational environment must be acknowledged.

First, a portion of the error variance is inherently driven by the human element of utility operations. Dispatch decisions are often influenced by field-level judgment calls, shifting priorities in the control room, or administrative nuances that are not captured by physical weather telemetry. Consequently, even a perfect meteorological model cannot account for every deviation in personnel assignment.

Second, the disparity in data resolution introduced unavoidable spatial noise. Because

SORT dispatch logs are aggregated at the ZIP code level while weather data is recorded at specific station coordinates, we engineered a 12-hour spatial proxy to link these sources within the H3 hexagonal grid. While this was the most robust method for aligning the datasets, this process of creating a proxy naturally introduces a minor degree of geographic uncertainty that can affect localized predictions.

Finally, the dashboard's comprehensiveness was constrained by data privacy and data security protocols. Due to the sensitive nature of utility infrastructure, certain granular details regarding asset vulnerability or specific grid configurations were omitted from the public-facing dashboard. These restrictions, while necessary for security, limit the level of contextual detail available to the end-user.

4.2 Future Work

Building upon the baseline established by PREVAIL, several high-impact avenues exist for future development.

- **Real-Time API Integration:** The most immediate enhancement would be the integration of a live weather API. By streaming real-time meteorological data directly into the inference pipeline, the model could generate dynamic, "on-the-fly" workforce forecasts as storms evolve, rather than relying on batch-processed historical telemetry.
- **Logistical Refinement:** The deployment pipeline could be further optimized by incorporating real-time traffic and road closure data. Integrating these variables would allow the model to adjust staging recommendations based on the actual travel time required for crews to reach an incident location during adverse conditions.
- **Multimodal Failure Prediction:** While the current framework focuses on personnel counts, the underlying Poisson architecture could be expanded to predict specific equipment failures, such as transformer blowouts versus vegetation-related line faults. Mapping specific hardware needs alongside crew sizes would provide a truly holistic logistical solution for emergency response.

5 Conclusion

The PREVAIL framework successfully demonstrates that the transition from reactive to proactive grid management is not only mathematically viable but operationally essential. By shifting the analytical focus from traditional outage probability to explicit resource quantification, this project provides a scalable template for utility operators to manage the increasing volatility of extreme weather. The stacking ensemble model proves that even within the complex, non-linear environment of storm response, machine learning can deliver reliable, localized logistics that bridge the gap between meteorological data and field-level action.

The broader impact of this work extends beyond operational efficiency; it directly affects community resilience. For utility providers like SDG&E, the ability to accurately stage crews before a weather event makes landfall means significantly reducing expensive standby contractor costs and, more importantly, minimizing the duration of power interruptions for critical infrastructure and residents. As extreme weather events continue to grow in frequency and intensity, tools like PREVAIL will be fundamental in ensuring that the energy grid remains a reliable backbone for society, exchanging the ”wait-and-see” approach of the past for a new data-driven standard for the future.

6 Acknowledgments

The development of the PREVAIL framework was made possible through the collaboration and support of several key individuals and teams. We would like to extend our sincere gratitude to our SDG&E mentors, Phi Nguyen, Mike Hilden, Chuck Hahm, and Fatemeh Aarabi, whose guidance was instrumental in shaping the project's operational focus.

We are also deeply grateful to Kasra Mohammadi, SDG&E's Weather Staff, and SDG&E's ERO Staff for providing the critical data and domain expertise necessary to approach this project. Your commitment to innovation has been a constant source of inspiration, and we are incredibly appreciative of the time and resources you invested in our success.

7 References

- Guikema, S. D., Nateghi, R., Quiring, S. M., Staid, A., Reilly, A. C., & Gao, M. (2014). Storm power outage prediction modeling. *Risk Analysis*, 34(10), 1998-2010.
- Nateghi, R., Guikema, S. D., & Quiring, S. M. (2011). Comparison and validation of statistical methods for predicting power outage durations in the event of hurricanes. *Risk Analysis*, 31(12), 1897-1906.
- Wanik, D. W., Anagnostou, E. N., Hartman, B. M., Frediani, M. E., & Astitha, M. (2015). Storm outage modeling for an electric distribution network in Northeastern USA. *Natural Hazards*, 79, 1359-1384.